MAC Protocols in Ad Hoc Networks

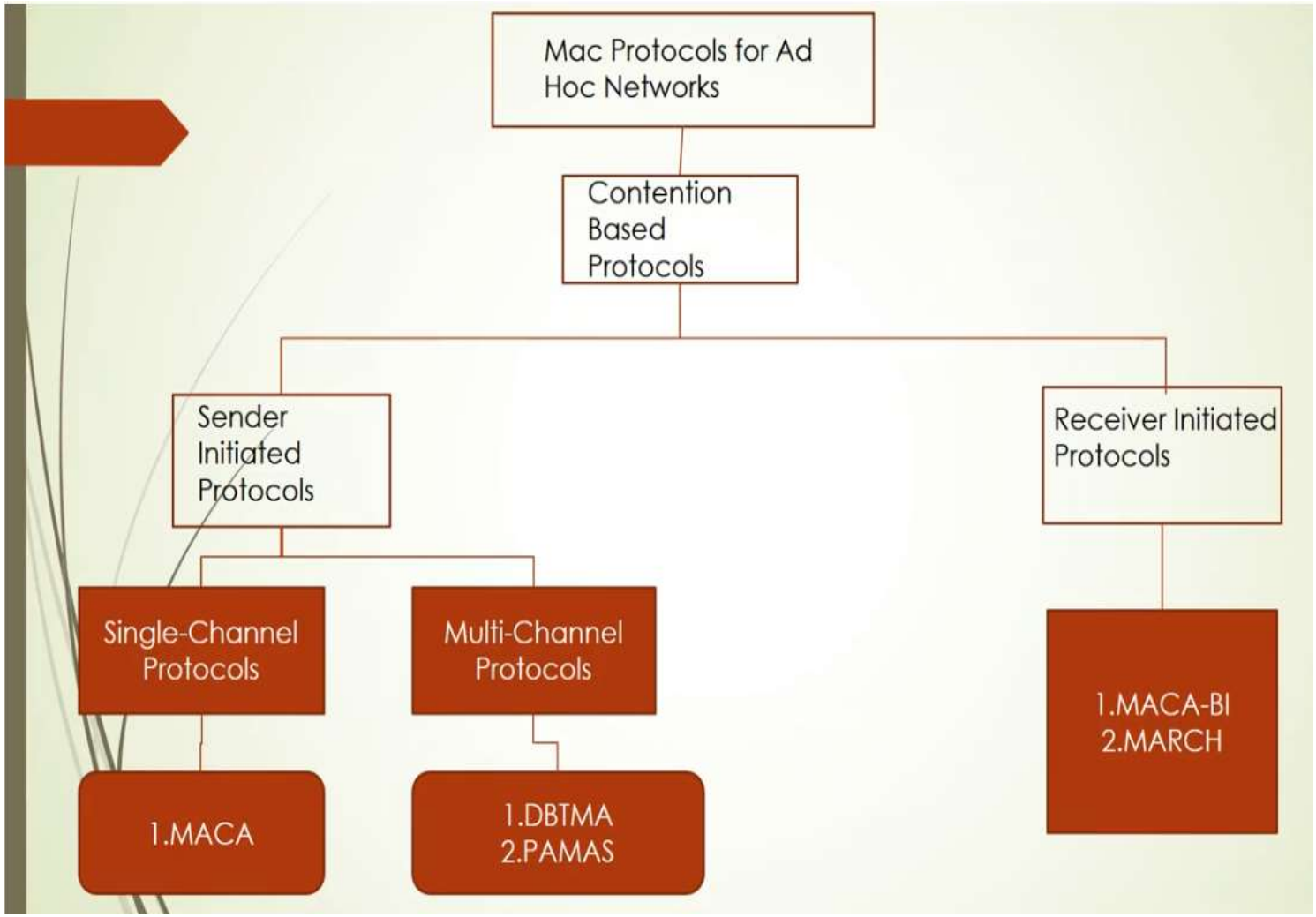
Types of MAC Protocols :-

1. Contention Based Protocols without any Reservation or Scheduling Mechanism.

2. Contention Based Protocol with Reservation Mechanism.

3. Contention Based Protocol with Scheduling Mechanism.

4. Other MAC Protocols


Mac Protocols for Ad Hoc Networks
Contention Based Protocols
Sender Initiated Protocols
Receiver Initiated Protocols
Single-Channel Protocols
Multi-Channel Protocols
1.MACA-BI
2.MARCH
1.MACA
1.DBTMA
2.PAMAS

Existing Ad Hoc MAC Protocols :-

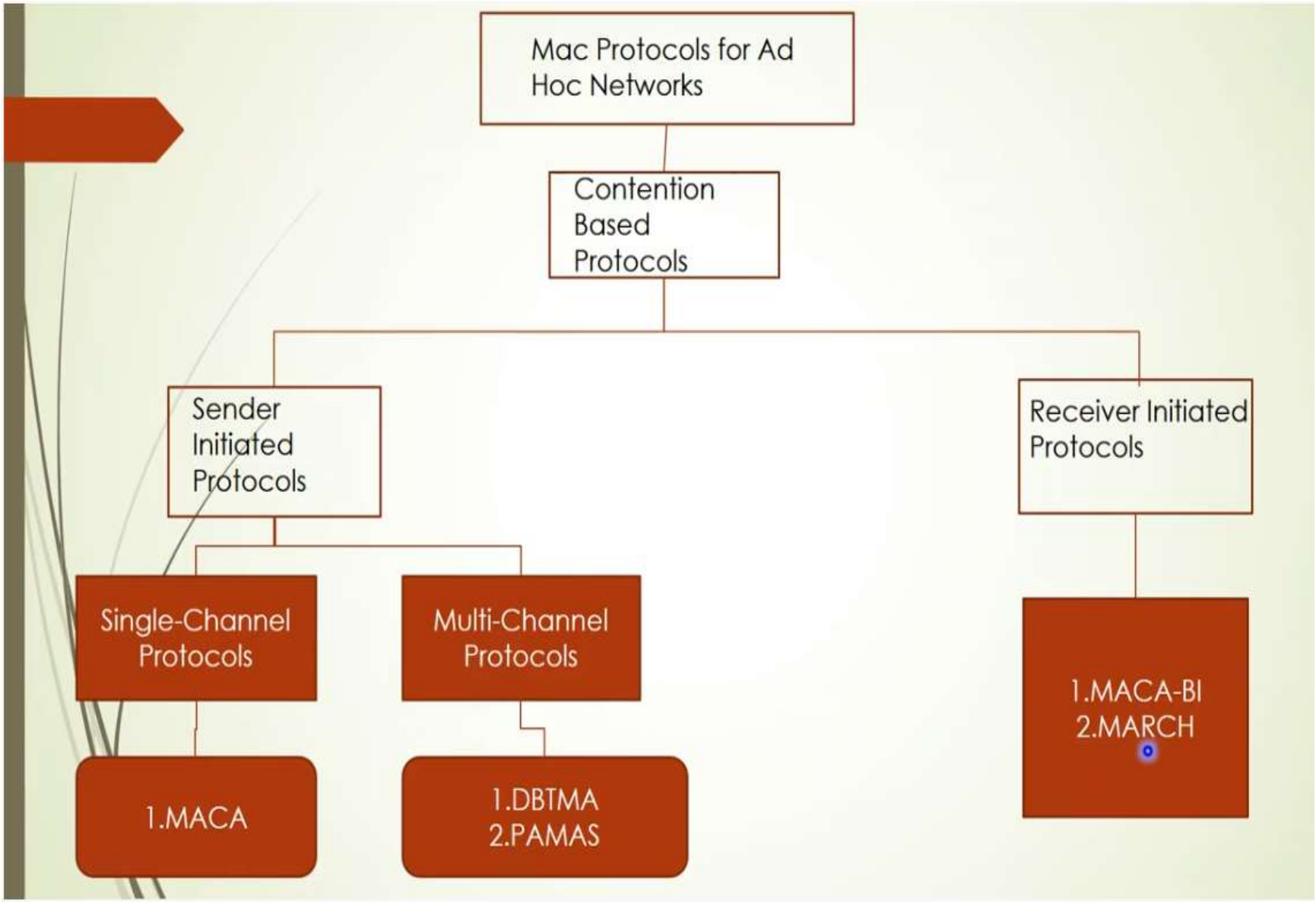
1.Multiple Access With Collision Avoidance **(MACA)**

2.Multiple Access With collision Avoidance By Invention **(MACA-BI)**

3.Power-Aware Multi-Access Protocol with Signaling **(PAMAS)**

4.Dual Busy Tone Multiple Access **(DBTMA)**

5.Media Access with Reduced Handshake **(MARCH)**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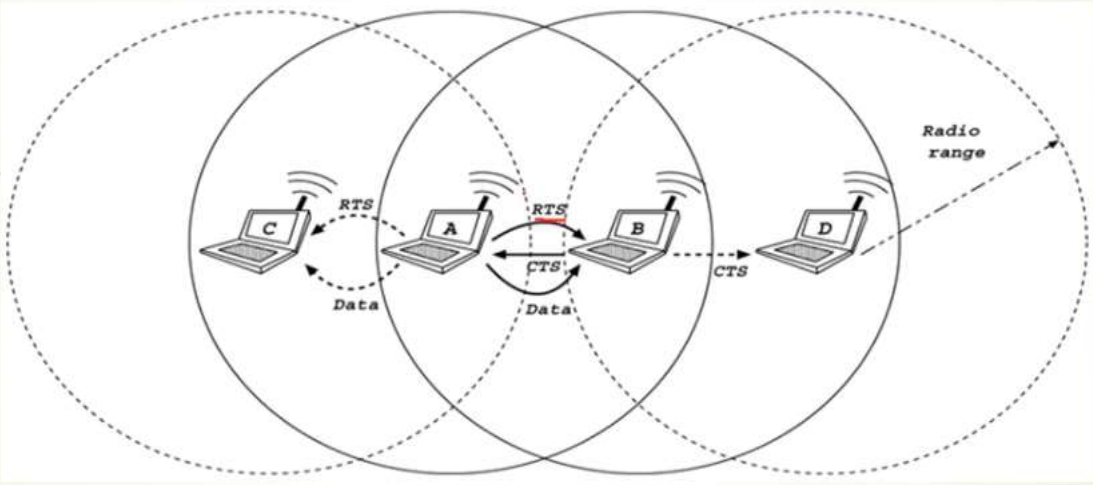
1. According to Initiation Mechanism-

1.MACA	Sender Initiated Protocol
2.MACA –BI	Receiver Initiated Protocol
3.PAMAS	Sender Initiated Protocol
4.DBTMA	Sender Initiated Protocol
5.MARCH	Sender and Receiver Initiated Protocol

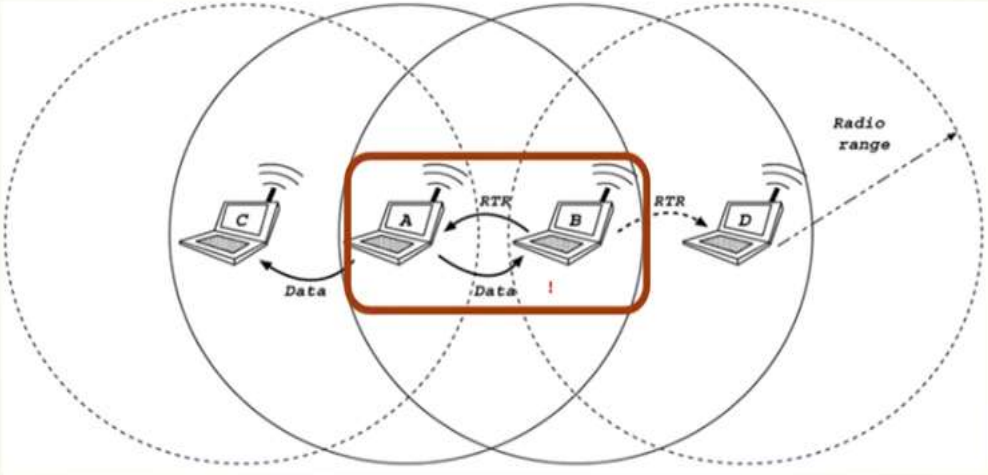
Sender-Initiated MAC Protocols



Receiver-Initiated MAC Protocols

The receiver (node B) first has to contact the sender (node A), informing the sender that it is ready to receive (RTR) data.

- A node cannot transmit data unless it has received an invitation from the receiver.



2. According to Channel Allocation :-

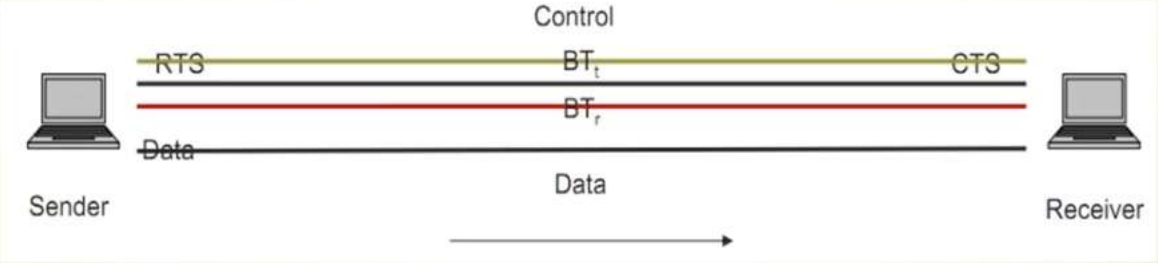
1.MACA	Single Channel Protocol A node that wins the contention to the channel can make use of the entire bandwidth.
1.MACA-BI	Single Channel Protocol
3.PAMAS	Multi-Channel Protocol 1.Signal Channel-control signals transfer over this Channel 2.Data Channel-Data signals transfer over this Channel
4.DBTMA	Multi-Channel Protocol Data Channel And control channel
5.MARCH	Single channel

3. According to Handshaking And Data transfer Mechanism

1.MACA	Uses a three-way handshake, RTS-CTS-Data.
1.MACA-BI	MACA-BI uses only a two-way handshake. (RTR-Data)
3.PAMAS	Uses a three-way handshake.(RTS-CTS-Data) But here Separate power on/off mechanism is present.
4.DBTMA	Busy tone mechanism is used in control channel. It uses three-way Handshaking (RTS-CTS-Data)
5.MARCH	It uses omni-directional antenna to reduce overhearing It uses three way handshaking(RTS-CTS-Data) only upto first hop. Then it is reduced to two-way handshake. (CTS-Data)

Busy Tone mechanism of **(DBTMA)**

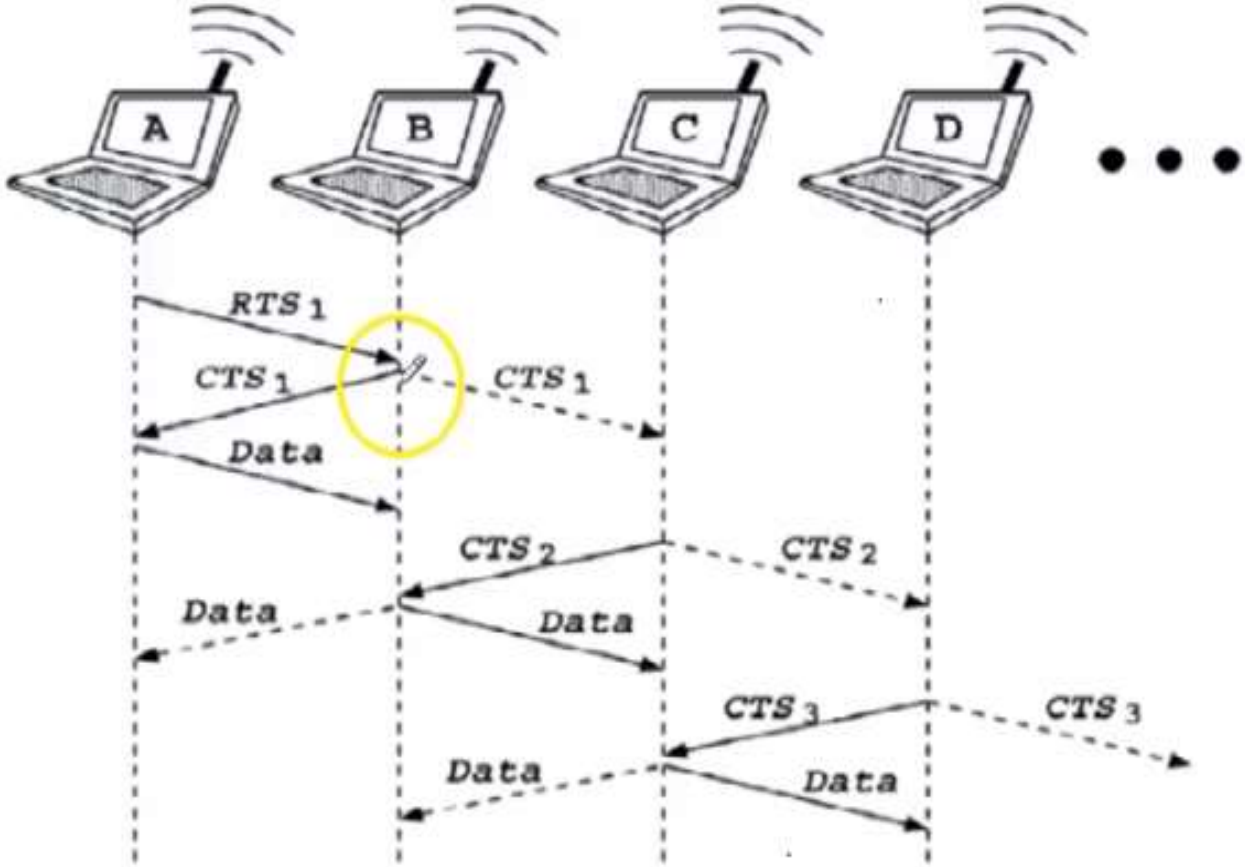
1. BTt -Transmit busy tone
2. BTr -Receive busy tone



Power Off mechanism of PAMAS:-

- PAMAS is based on the **MACA protocol** with the addition of a separate **signalling channel.**
- PAMAS conserves battery power by selectively **powering off nodes** that are not actively transmitting or receiving packets.
- In PAMAS, nodes are required to shut themselves off if they are overhearing other transmissions not directed to them.
- If a node has no packets to transmit, it should power off if one of its neighboring nodes is transmitting.
- If a node has packets to transmit, but at least one of the neighboring nodes is transmitting and another is receiving, then it should power off its transceiver.

MARCH Protocol



MARCH :Media Access with ReduCed Handshake

- MARCH exploits the overhearing characteristic associated with an ad hoc mobile network employing omni-directional antenna.

- MARCH is a sender and receiver initiated protocol

- MARCH supports fast data transfer over a multi-hop route

- MARCH provides data flow control by using a sequence number contained inside the CTS packet

- Improves communication throughput in wireless multihop Ad-hoc networks by reducing the amount of control overhead.

•For an ad hoc route of n hops, the number of handshakes needed to send a data packet from the source to the destination is only (n + 1) in MARCH.

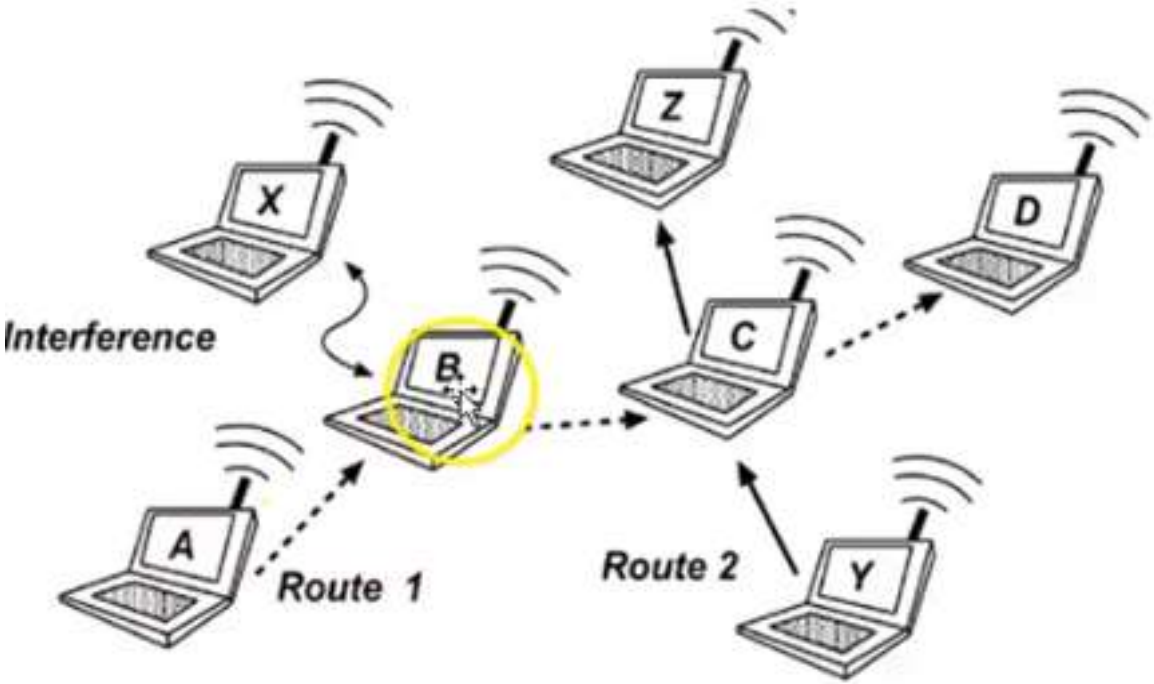
• There are n −1 intermediary nodes between the source and destination.

•MARCH can be viewed as a request-first, pull-later protocol since the subsequent nodes in the path just need to send invitations to pull the data toward the destination node.

▪ The RTS-CTS message in MARCH contains:

1. The MAC addresses of the sender and receiver
2. The route identification number (RTID)

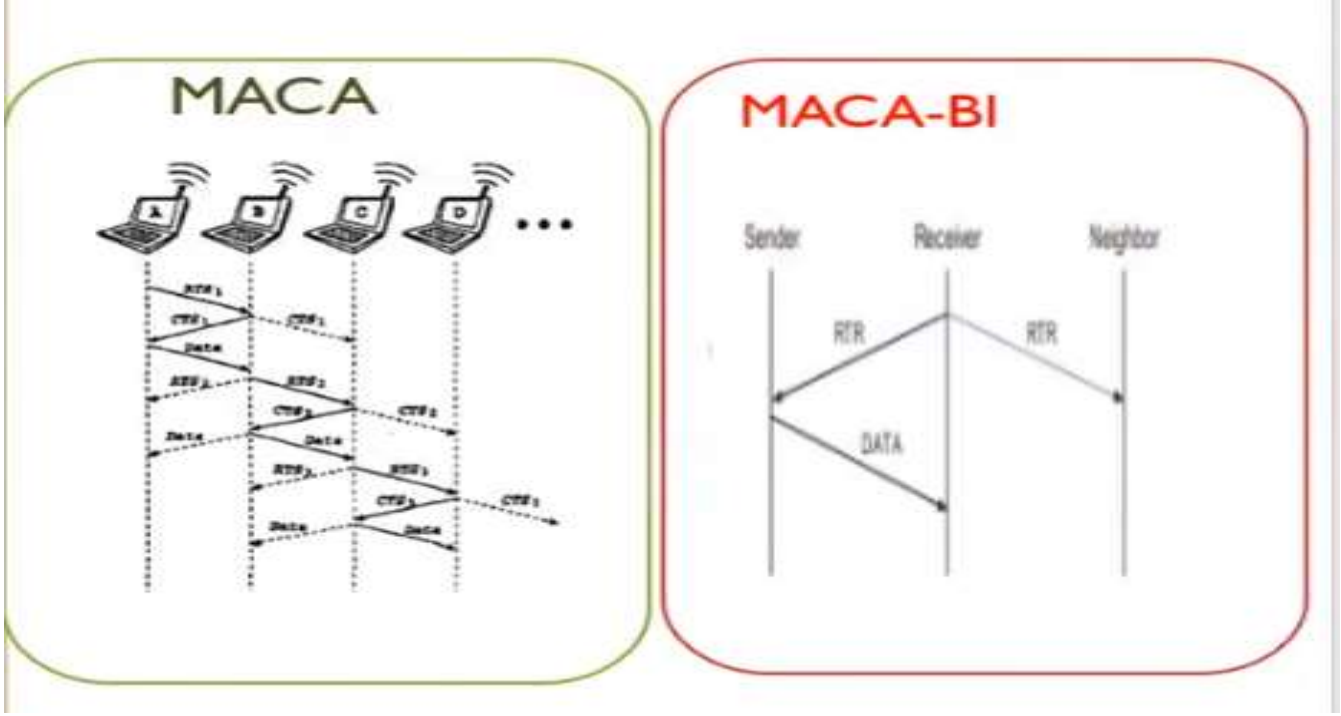
Application of MARCH:

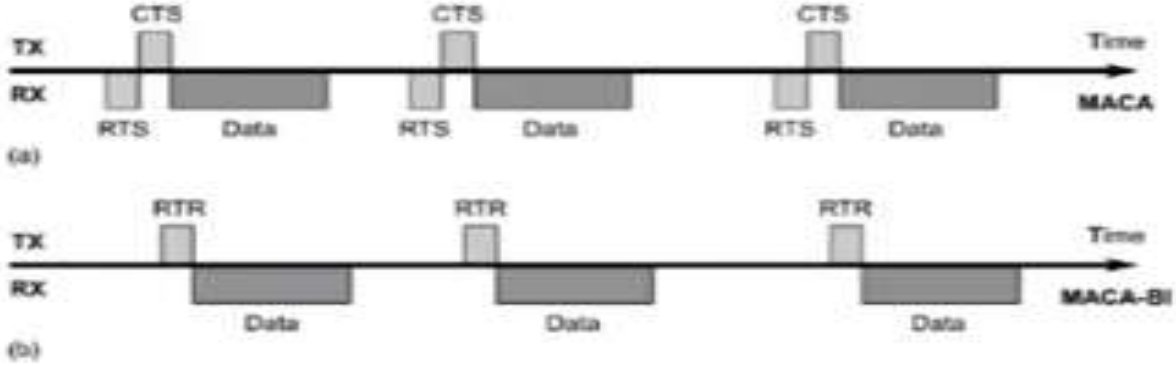


MACA	Vs	MACA-BI
1. Sender initiated MAC Protocol		1. Receiver initiated MAC Protocol
2. Ready-to-Send (RTS) and Clear-to-Send (CTS) is used.		2. RTR(Ready To Receive) is used.
3. Three-way handshake.		3. Two-way handshake.
4. RTS-CTS-DATA		4. RTR-DATA
5. Handshakes needed to send a data packet from the source to the destination is 2n ,where n is hopes.		5. Handshakes needed to recive a data packet from the source to the destination is n ,where n is hopes.
6. efficiency is low.		6. efficiency is high.
7. Receiver no needs to predict if node has data to transmit to it.		7. Receiver needs to predict if node has data to transmit to it.
8. Overhead more.		8. Overhead less.
9. Speed is less .		9. Speed is more.

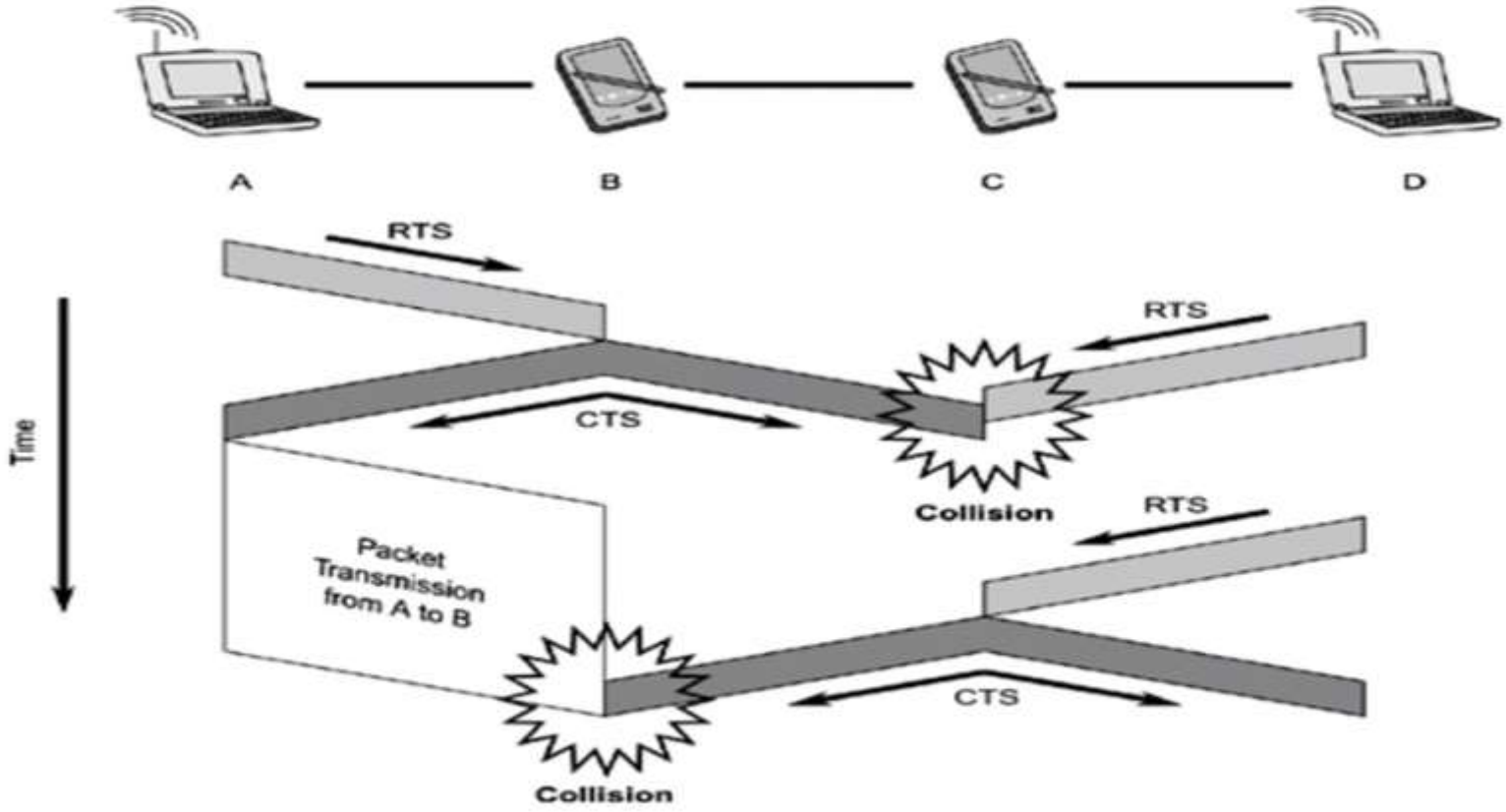
4. According to Number of Handshakes needed to send a data packet from the source to the destination for the Route of N hops:-

MACA	2N Handshakes
MACA-BI	N Handshakes
MARCH	N+1 Handshakes

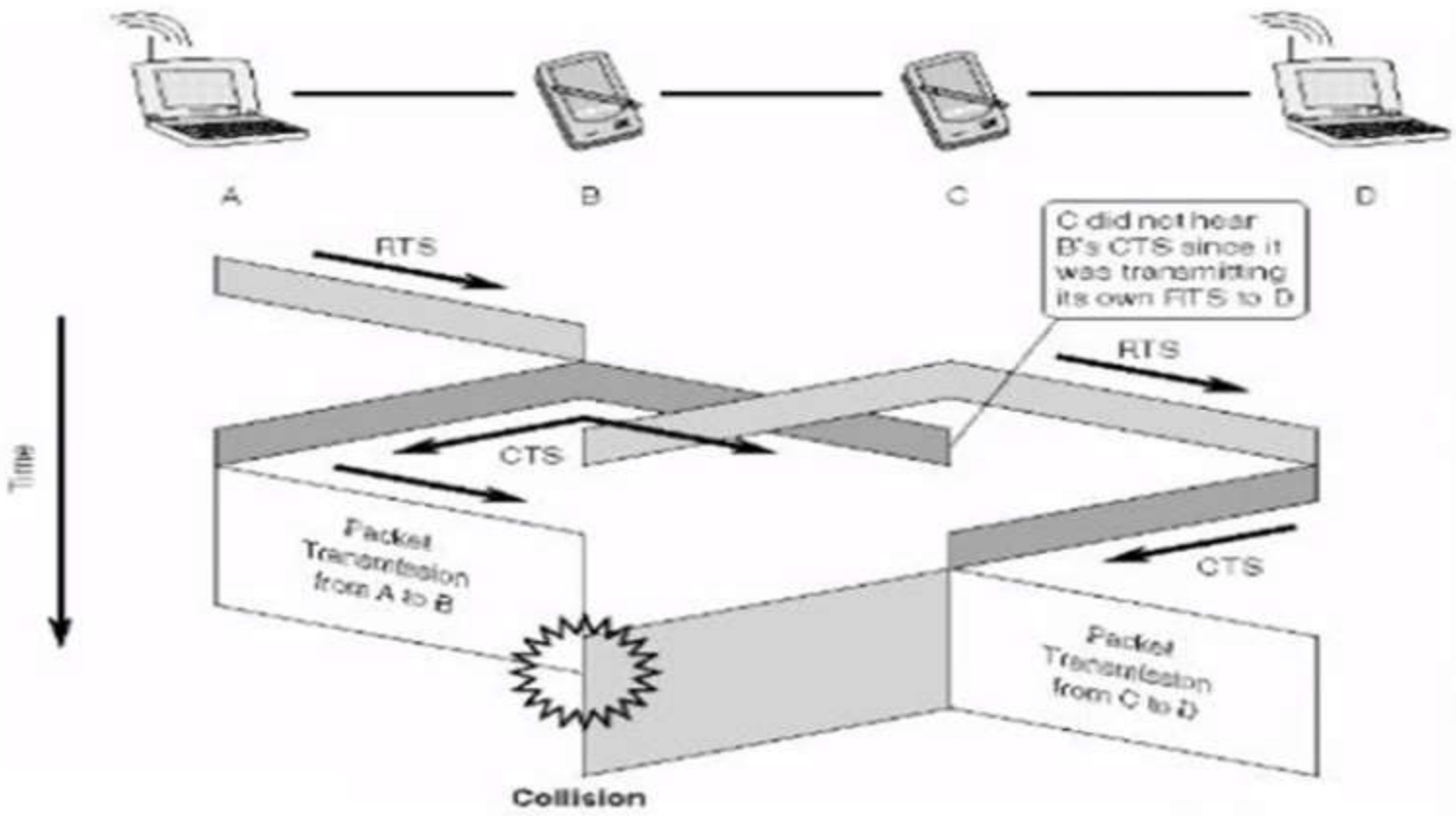

MACA
A
B
C
D
MACA-BI
Sender
Receiver
Neighbor
RTR
RTR
DATA


CTS
CTS
CTS
TX
RX
Time
MACA
RTS
Data
RTS
Data
RTS
Data
(a)
RTR
RTR
RTR
TX
RX
Time
MACA-BI
Data
Data
Data
(b)

Collision


A
B
C
D
RTS
RTS
CTS
Collision
RTS
Packet
Transmission
from A to B
CTS
Collision
Time

Collision


A
B
C
D
RTS
C did not hear B's CTS since it was transmitting its own RTS to D
RTS
CTS
Time
Packet Transmission from A to B
CTS
Packet Transmission from C to D
Collision